# Graph-Based Propagation for Multispectral Remote Sensing Image Completion

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***Abstract*— The image completion refers to the problem of recovering the missing, corrupted, or obscured entries in image data. In this article, we consider the problem in the remote sensing domain, where regions of an image are missing due to difficulties such as cloud cover, sensor failures, or partial sensor coverage. Where the previous work in this field generally falls into the category of low-rank completion methods, we propose a novel graph-based diffusion approach to the problem. The method, referred to as GraphProp, propagates observed entries around a graph-based representation of the image region in order to recover the missing entries. The graph-based diffusion approach to completion is to the best of the authors' knowledge a novel method for remote sensing image completion. Using real-world multispectral image data acquired from the Landsat 7 platform, we validate our approach using experiments which synthetically obscure image sections. In these tests, we benchmark against alternative image completion approaches and demonstrate the superior reconstruction performance of our method versus the state of the art. The code which implements the method has been made publicly available at: https://github.com/iainrolland/GraphProp.**

***Index Terms*— Graph propagation, graph theory, image completion, missing data, remote sensing, tensor completion.**

## I. Introduction

REMOTELY sensed observations offer an insight into a great number of different natural and human processes [1]. The ability of Earth observation satellites to take measurements with global coverage is one of their great strengths. However, there are many occurrences which can prevent complete coverage [2], such as sensor failures [3], [4], cloud obscuration [5], [6], and irregularly spaced revisit times. In addition to this, there are many scenarios which benefit from considering multiple data sources [7], [8], [9]. This provides an additional dimension of data which is often incomplete due to the differences in orbits leading to partially overlapping measurements.

In order to have the most information from which to extract insights, one would ideally have access to observations at all locations of interest at all times. In reality, this is very rarely possible, and instead, the available acquisitions are often only partially observed. The problem of estimating the missing image entries describes what can be referred to as the image completion problem.

Two dominant approaches for image completion exist in the remote sensing domain: low-rank completion methods [10], [11], [12] and deep learning approaches [13], [14], [15]. Low-rank completion refers to the family of methods which consider the partially observed image as a matrix or tensor and assume that the data lies in some low-rank subspace. These methods aim to exploit this low-rank assumption in order to complete the missing entries based on observed entries. Although the real-world data are never perfectly low rank, the distribution of singular values can be examined to show that the majority of singular values are of much smaller magnitude than the handful of largest singular values. Using this fact to argue that the low-rank assumption is valid, algorithms have been presented [16], [17], [18] which complete the matrix or tensor using the values that provide the lowest rank solution satisfying the constraints on the observed entries. While low-rank completion methods perform impressively for data which is missing at random, it cannot perform the completion of any column or row which is missing in its entirety [19]. Although this represents an extreme case, it is not uncommon for the missing patterns in real-world completion problems to form large contiguous patches which approach this scenario. The majority of existing theoretical work investigating the extent to which tensors are recoverable from partial observations is based on the assumption that observations are made uniformly at random. This, in reality, represents only a tiny fraction of remote sensing image completion applications. Many applications, for example, shadow correction [20], sensor failures [3], [4], and cloud removal [7], [21], [22], feature missing patterns which could not be considered uniformly random.

The second dominant approach for remote sensing image completion is methods which make use of deep learning architectures. This term is used to encompass a wide range of approaches and network architectures but might refer to the literature which makes use of any or a combination of generative adversarial networks (GANs) [23], convolutional neural networks, and encoder–decoder approaches [24]. These

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methods again have shown in various contexts to have their merits but they all require large volumes of data and computational resources in the training process. While this is not a significant barrier for certain applications, it is not always practical or efficient to take this approach. We argue that there remains a significant demand for image completion methods which can be applied to any problem or sensor without the upfront investment required to train a deep learning architecture to the specifics of the task.

In order to fulfill a need for a remote sensing image completion method which can handle large contiguous sections of missing entries as well as quickly and efficiently generalize across any sensor or measurement without training data, we propose a graph-based diffusion algorithm. Taking a graph and a partially observed image as inputs, our approach propagates values around a graph-based representation of the image in order to fill the missing entries. This can be performed efficiently by solving a set of linear equations to obtain an analytical solution for the system's steady-state solution or in an iterative fashion until convergence is reached. For this method, we derive an upper bound on the completion error and relate this to an existing upper bound derived for an alternative graph signal recovery method in order to validate our approach. The experimental testing is conducted using a synthetically obscured remote sensing dataset, and the performance is benchmarked against the state-of-the-art low-rank completion and graph signal recovery methods. In our analysis, we show that the proposed method, referred to as GraphProp, can outperform state-of-the-art methods and strategies both in quality of completion and in computational efficiency.

The contributions of this article are summarized as follows. Existing image completion methods which rely on low-rank completion are not designed to handle the large contiguous regions of missing data that often occur in remote sensing images. Therefore, a new approach has been developed to tackle this problem. The graph-based representation has been developed so as to describe the region's similarities using connections portrayed by graph edges. Completion of the partially observed data can then be performed by exploiting equations of graph-based diffusion. The graph-based approach to remote sensing image completion is to the best of the authors' knowledge a novel solution to the problem.

The rest of this article is organized as follows. Section II introduces and reviews the existing literature in remote sensing image completion. Section III gives details of our proposed approach and presents the GraphProp algorithm. Section IV derives an upper bound on the completion error and relates this to an existing upper bound in the literature for graph-based signal recovery. Section V presents a series of experiments involving synthetically obscured remote sensing data and makes comparisons with the state-of-the-art methods. Finally, Section VI provides an overview of the contribution and concludes with potential directions of the future research.

## II. Related Works

In this section, a review of existing methods for remote sensing image completion is presented, with a summary of the strengths and weaknesses of each approach.

### A. Low-Rank Completion Methods

The ill-posed problem of recovering a matrix which is only partially observed was shown by Candès and Recht [25] to in fact be possible, under the condition that the original matrix was low rank. In many cases, the partially observed matrix can be used to recover the original data exactly. The family of methods which can be described as low-rank completion algorithms aims to exploit this low-rank property to reconstruct the missing entries from the observations. To perform this completion, they express an optimization problem which looks to minimize the rank of the completed matrix subject to equality constraints on the entries which are observed. Minimization of the rank directly is NP-hard, so instead the nuclear norm, which is the convex relaxation of rank, is minimized. An algorithm for completion of this type was originally proposed for matrix completion [26] and later generalized to tensors by Liu et al. [27]. There is extensive literature [17], [18], [28], [29], [30] focusing on developing better algorithms for tensor completion which are capable of improved-quality reconstructions and expansion of the limits of recoverability in terms of minimum required observations. The theoretical limits of such approaches have also been comprehensively studied by investigations such as [31], [32], and [33] which provide lower bounds on completability.

### B. Deep Learning Approaches

In remote sensing, missing data completion tasks often present themselves in the form of a cloud removal task [7], [21], [22]. These tasks are often tackled using deep learning approaches. To facilitate the training of such models datasets such as the SEN12MS dataset [34] have been created, a dataset which consists of image triplets: cloud-free multispectral Sentinel-2 images, Sentinel-1 synthetic aperture radar (SAR) images, and moderate resolution imaging spectroradiometer (MODIS) land cover maps. This dataset was augmented by Meraner et al. [21] to include cloudy Sentinel-2 images into a dataset named SEN12MS-CR (cloud removal). Meraner et al. [21] use this dataset to train a ResNet architecture which takes as input a cloudy Sentinel-2 image and a Sentinel-1 SAR image and is optimized using a loss computed against the reference cloud-free Sentinel-2 image. The dataset consists of 157 521 patch triplets of size $256 \times 256 \times 28$ and occupies 620 GB of disk space. Their model, named DSen2-CR, is robust even in examples of almost complete cloud cover and compares superiorly to pix2pix [35], a GAN-based approach using a U-Net generator.

Although deep learning approaches have been shown capable of producing high-accuracy results, they have been shown to suffer from a concept referred to as artifact generation [21], which is when the model makes changes to the sections of the image which were not obscured. There are techniques which have been proposed which aim to minimize this effect but they do not eliminate the artifacts entirely. In addition to this, the accuracy of such methods has been shown highly correlated with the size of the available training dataset [36], [37]. In computer vision, the state-of-the-art results are often working with datasets which contain millions of training

images [38], and there is a general consensus in the field that deep learning architectures require large volumes of training data in order to generalize [39]. In most applications, a large and well-organized dataset suited to the specifics of the completion problem will not already exist, and it is not always practical to curate such a dataset from scratch.

In the many applications for which the upfront investment to train a high-accuracy deep learning model is not practical, there is a need for improved-accuracy completion methods which generalize quickly and do not require a training dataset.

### C. Graph-Signal Recovery

Although not generally used to represent and recover images as graphs, the idea of graph-based signal recovery is not new [40], [41], [42], [43]. The graph-signal recovery can be thought of as a subset of the wider and well-studied problem of signal recover in the field of signal processing. In any problem which involves signal recovery relating to data which can be considered to lie on a graph, the problem can be referred to as the graph signal recovery. Chen et al. [44] propose the use of the graph shift operator to minimize total variation in order to perform graph signal inpainting. This idea is extended in [45] which generalizes the idea across a number of problems relating to recovering signals on graphs, one of which is the signal inpainting problem. The assumption which they make in order to perform signal recovery is that the graph signal is smooth as quantified by the measure of graph total variation. Expressing the problem as a minimization of graph total variation with equality constraints on the observed entries, they perform completion by finding the analytical optimum of this minimization problem. The approach assumes knowledge of a graph structure a priori and normalizes the adjacency matrix such that the largest eigenvalue is equal to 1. Naming this approach graph signal inpainting via total variation minimization (GTVM), they proceed to derive an upper bound on the error of the completion. Although we draw analogies between our approach and GTVM, we do not make the same assumption of smoothness which we consider an over-simplification of the problem when representing image data as a graph. Our quantitative results show that our approach significantly outperforms GTVM in terms of error.

## III. Methods

### A. Graph Theory

Consider a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ which is composed of $|\mathcal{V}| = n$ vertices $v_1, \ldots, v_n$. The graph's edges can be represented by an adjacency matrix, $\mathbf{A} \in \mathbb{R}^{n \times n}$, with entries given by

$$A_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

in the unweighted case or if the graph's edges are weighted then $A_{ij}$ is equal to the weight of $(v_i, v_j) \in \mathcal{E}$. In this analysis, we will consider strictly undirected graphs, which is to say $A_{ij} = A_{ji}$. For the avoidance of doubt, we state here that $A_{ii} = 0, \forall v_i \in \mathcal{V}$, which is to say we use the adjacency matrix without self-connections.

It is also useful to define two further quantities which derive from the adjacency matrix: the degree matrix, $\mathbf{D}$, and the unnormalized graph Laplacian, $\mathbf{L}$. The degree matrix is a diagonal matrix in $\mathbb{R}^{n \times n}$ with diagonal entries given by $D_{ii} = \sum_j A_{ij}$. The unnormalized graph Laplacian is given by $\mathbf{D} - \mathbf{A}$ and, therefore, is also in $\mathbb{R}^{n \times n}$.

### B. Images as Graphs

In this approach, we consider a graph-based representation of our image data. Often, in the context of a multiband image, we are working with a third-order tensor, with two spatial dimensions and one spectral dimension. We will consider each spatial location as a graph node and the location's spectral signature as the node's feature vector. Therefore, an image of shape $A \times B \times d$, representing height, width, and number of bands, respectively, can be represented as a graph with $A \times B$ nodes and $d$ features per node. In order to connect our nodes into a graph, we will make use of a reference image. This is an approach which is well-suited to problems in remote sensing, where we often have access to a co-registered reference image by virtue of an earlier acquisition. Considering the $A \times B$ nodes of the reference acquisition in $\mathbb{R}^d$, we compute the $k$-nearest neighbors of each node and connect any two nodes which belong to either node's $k$-nearest neighbors. This graph provides our adjacency matrix to be used to recover the missing image entries.

The proposed approach of exploiting a previously acquired image of the region is considered practical thanks to readily available repeat imagery provided by modern remote sensing missions [46], [47], [48]. The Sentinel-2 mission, for example, provides a 5-day revisit cycle [49], whereas an 8-day revisit is made by one of either Landsat-8 or Landsat-9 [50]. Given that, as depicted in Fig. 1, GraphProp can be applied to a series of partially observed images of the region, the algorithm does not necessarily require cloud-free inputs. Therefore, it is highly likely that given a partially observed image to complete, a recently acquired image of the same region will be available that allows GraphProp to be applied. The only other input the algorithm requires is a mask indicating the locations of the missing entries. This is also easily obtained in practice through the use of the quality assurance (QA) band which is commonly provided alongside remote sensing data [49], [51].

### C. Graph-Based Diffusion

Motivated by the idea that a graph structure, since graphs can be used to represent the similarities within a dataset, could act as a mechanism along which information can be disseminated, we adopt graph-based diffusion to perform completion. This method, represented schematically in Fig. 1, aims to exploit the information contained within the observed entries and harnesses graph-based diffusion to propagate this information to the rest of the graph nodes. In order to obtain a graph-based diffusion scheme, the generalization of the 1-D heat diffusion equation to a graph-based diffusion equation is presented.

The 1-D diffusion of a scalar $h$ is represented by the partial differential equation

$$\frac{\partial h}{\partial t} = \alpha \frac{\partial^2 h}{\partial x^2} \tag{2}$$

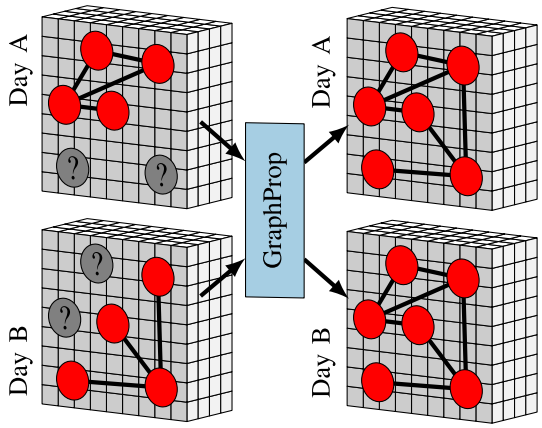



Fig. 1. Graph nodes are used to represent spatial locations and graph edges are used to connect spectrally similar pixels. The observed regions are used to obtain the graph's adjacency matrix. The graph structure is used to propagate observed entries in (left) inputs such that the missing entries are reconstructed in (right) outputs.

where $t$ and $x$ represent the temporal and spatial variables, respectively, and $\alpha$ represents the diffusivity constant [52]. Through the use of truncated Taylor series expansions, an update equation for the system can be obtained as follows:

$$h_{t+1}^{x} \approx h_{t}^{x} + \frac{\alpha \Delta t}{\Delta x^2}\left(h_{t}^{x-1} - 2h_{t}^{x} + h_{t}^{x+1}\right) \tag{3}$$

where $\Delta t$ and $\Delta x$ are the sizes of the discretization in time and space, respectively.

The numerical scheme can be easily extended to reflect boundary conditions [53]. For example, if we require a value at a location $x$ to be constant through time, that is, $\partial h_t^x/\partial t = 0, \forall t$, we apply the boundary condition simply by holding the value as constant at these locations during the update step.

This approach is easily related to diffusion on a graph by noticing that the term $(h_t^{x-1} - 2h_t^x + h_t^{x+1})$ can be generalized to beyond the 1-D as a summation of the features at the neighboring nodes and a subtraction of the node's own feature multiplied by its degree. By stacking a scalar $h$ for each node into a vector, $\boldsymbol{h} \in \mathbb{R}^n$ is obtained. In the context of multispectral remote sensing data, the vector $\boldsymbol{h}$ is used to denote one spectral channel, where values in the vector denote each pixel's value in the corresponding band. The update equation for graph diffusion [54], [55] can therefore be expressed as follows:

$$\frac{\partial \boldsymbol{h}}{\partial t} \propto (\mathbf{A} - \mathbf{D})\boldsymbol{h} = -\mathbf{L}\boldsymbol{h} \tag{4}$$

where $\mathbf{A}$, $\mathbf{D}$, and $\mathbf{L}$ are the graph adjacency, degree, and unnormalised Laplacian matrices, respectively, as introduced in Section III-A.

This equation can be used to propagate the features from the observed nodes to the unobserved nodes. In our approach, we consider the image completion as the steady-state solution obtained by letting $t \to \infty$ under the boundary condition that the observed nodes' features are held constant.

The motivation for providing the image completion solution as the steady-state solution to the described diffusion is as follows. When diffusion around the graph is performed, the features at nodes which were not observed are updated according to influences from their neighbors' feature values. The longer diffusion is allowed to continue, the more similar to their graph neighbors the values at each node become. By virtue of the graph having been constructed to connect image pixels which have exhibited spectral similarities, this convergence toward similarity is desirable. Given that similarities improve as diffusion continues, the best solution is therefore considered to be the state that is reached when diffusion is complete, denoted as the steady-state solution.

This solution can be found either by iteratively applying small update steps proportional to the temporal derivative or alternatively by considering that the steady-state solution is when the temporal derivative is 0. The final solution does not depend on the initial conditions given for the unobserved entries in $\boldsymbol{h}$. To write the analytical solution neatly, we will borrow the notation of [45] and use $\mathcal{U}$ and $\mathcal{M}$ to denote sets containing the indices of the unobserved and observed nodes, respectively. These sets are of size $|\mathcal{U}| = n_{\mathcal{U}}$ and $|\mathcal{M}| = n_{\mathcal{M}}$, denoting the number of unobserved nodes and observed nodes, respectively. Each node must exist in either $\mathcal{U}$ or $\mathcal{M}$ but not both, such that $n = n_{\mathcal{U}} + n_{\mathcal{M}}$. Where these sets are used as subscripts, we denote that specific rows or columns have been indexed corresponding to the nodes of the respective set. Without loss of generality, we can reorder the nodes in the graph such that the observed nodes come first and the unobserved nodes afterward. This allows us to express the boundary conditions which hold observed features constant, $(\partial \boldsymbol{h}_{\mathcal{M}}/\partial t) = \mathbf{0}$, as follows:

$$\frac{\partial \boldsymbol{h}}{\partial t} = \begin{bmatrix} \mathbf{0} \\ \dfrac{\partial \boldsymbol{h}_{\mathcal{U}}}{\partial t} \end{bmatrix}. \tag{5}$$

By setting $(\partial \boldsymbol{h}_{\mathcal{M}}/\partial t) = \mathbf{0}$, the algorithm output is assured to have exact correspondence with the measured inputs provided for the observed nodes.

To satisfy (5) within (4), we set blocks within the matrix $\mathbf{L}$ to 0 to get

$$\frac{\partial \boldsymbol{h}}{\partial t} \propto \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ -\mathbf{L}_{\mathcal{U}\mathcal{M}} & -\mathbf{L}_{\mathcal{U}\mathcal{U}} \end{bmatrix} \begin{bmatrix} \boldsymbol{h}_{\mathcal{M}} \\ \boldsymbol{h}_{\mathcal{U}} \end{bmatrix}. \tag{6}$$

The steady-state solution is then found by solving for when the temporal derivative in (6) is equal to 0, such that $\boldsymbol{h}_{\mathcal{U}}$ can be found as the solution to the linear equations

$$\mathbf{L}_{\mathcal{U}\mathcal{U}}\boldsymbol{h}_{\mathcal{U}} = -\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{h}_{\mathcal{M}}. \tag{7}$$

When each node-based representation in the graph corresponds to multiple spectral bands we can stack each band to get a feature matrix, $\mathbf{H} \in \mathbb{R}^{n \times d}$. The feature matrix is formed according to

$$\mathbf{H} = \begin{bmatrix} | & & | \\ \boldsymbol{h}_1 & \cdots & \boldsymbol{h}_d \\ | & & | \end{bmatrix} \tag{8}$$

where $\boldsymbol{h}_d$ denotes the node features containing the $d$th spectral band.

This notation allows the method to be summarized within the pseudocode presented in Algorithms 1 and 2, where

**Algorithm 1** GraphProp Algorithm (SLC-Off Experiment)

**Input:** $\mathbf{H}^{(1)}, \mathcal{M}^{(2)}, \mathcal{U}^{(2)}, \mathbf{H}^{(2)}_{\mathcal{M}^{(2)}}$

$\mathcal{E} \leftarrow \text{kNN}\left(\mathbf{H}^{(1)}\right)$ ▷ Compute $k$-nearest neighbors graph

$\mathbf{L} = \text{Laplacian}(\mathcal{E})$

$\boldsymbol{H}^{(2)}_{\mathcal{U}} \leftarrow \text{Solve}\left(\mathbf{L}_{\mathcal{U}^{(2)}\mathcal{U}^{(2)}} \boldsymbol{H}^{(2)}_{\mathcal{U}^{(2)}} = -\mathbf{L}_{\mathcal{U}^{(2)}\mathcal{M}^{(2)}} \boldsymbol{H}^{(2)}_{\mathcal{M}^{(2)}}\right)$ ▷ Solve for steady-state solution of diffusion

**Output:** $\mathbf{H}^{(2)}_{\mathcal{U}^{(2)}}$

**Algorithm 2** GraphProp Algorithm (Partial Overlap Experiment)

**Input:** $\mathcal{M}^{(1)}, \mathcal{U}^{(1)}, \mathbf{H}^{(1)}_{\mathcal{M}^{(1)}}, \mathcal{M}^{(2)}, \mathcal{U}^{(2)}, \mathbf{H}^{(2)}_{\mathcal{M}^{(2)}}$

**for** each acquisition $\tau = 1, 2$ **do**

  $\mathcal{E}^{(\tau)} \leftarrow \text{kNN}\left(\mathbf{H}^{(\tau)}_{\mathcal{M}^{(\tau)}}\right)$ ▷ Compute $k$-nearest neighbors graph from observed pixels

**end for**

$\mathcal{E} = \mathcal{E}^{(1)} \cup \mathcal{E}^{(2)}$ ▷ Combine graphs

$\mathbf{L} = \text{Laplacian}(\mathcal{E})$

**for** each acquisition $\tau = 1, 2$ **do**

  $\boldsymbol{H}^{(\tau)}_{\mathcal{U}} \leftarrow \text{Solve}\left(\mathbf{L}_{\mathcal{U}^{(\tau)}\mathcal{U}^{(\tau)}} \boldsymbol{H}^{(\tau)}_{\mathcal{U}^{(\tau)}} = -\mathbf{L}_{\mathcal{U}^{(\tau)}\mathcal{M}^{(\tau)}} \boldsymbol{H}^{(\tau)}_{\mathcal{M}^{(\tau)}}\right)$ ▷ Solve for steady-state solution of diffusion

**end for**

**Output:** $\mathbf{H}^{(1)}_{\mathcal{U}^{(1)}}, \mathbf{H}^{(2)}_{\mathcal{U}^{(2)}}$

superscripts such as $\square^{(1)}$ and $\square^{(2)}$ denote the quantity relates to the first or second acquisition, respectively.

## IV. Error Upper Bound Derivation

In this section, we will derive an upper bound on the estimation error for the steady-state solution of the graph diffusion equation in (7). To do so, we will follow an approach taken in [44] and [45], where they consider a graph-based optimization subject to the equality constraints provided by the observed entries. We will first, therefore, rephrase our method in terms of an optimization problem analogous to that shown in [44] and [45].

Considering an optimization with respect to $\boldsymbol{h}$, we dictate that $\boldsymbol{h}_{\mathcal{M}} = \boldsymbol{r}_{\mathcal{M}}$, where $\boldsymbol{r}_{\mathcal{M}}$ is the measured entries of the observed nodes, in order to ensure the final solution matches the observations. The following optimization:

$$\begin{aligned} &\min_{\boldsymbol{h}} \; ||\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{h}_{\mathcal{M}} + \mathbf{L}_{\mathcal{U}\mathcal{U}}\boldsymbol{h}_{\mathcal{U}}||_2^2 \\ &\text{s.t. } \boldsymbol{h}_{\mathcal{M}} = \boldsymbol{r}_{\mathcal{M}} \end{aligned} \tag{9}$$

can be seen by inspection to possess a closed-form solution matching the result given in (7).

To allow us to follow the derivation process in [44] and [45], we must rephrase this optimization problem slightly. By again ordering observed nodes to come before unobserved nodes, we can consider block matrices which use $\mathcal{M}$ or $\mathcal{U}$ to refer to rows and columns depending on whether they correspond to observed or missing nodes, respectively. Matrices $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$, defined in Table I, are used in this section to ease exposition. For example, matrix $\mathbf{P}$ represents a block matrix given by

$$\begin{aligned} \mathbf{P} &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} (\mathbf{I} - \mathbf{D}^{-1}\mathbf{A}) \\ &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ -\mathbf{D}^{-1}_{\mathcal{U}\mathcal{U}}\mathbf{A}_{\mathcal{U}\mathcal{M}} & \mathbf{I} - \mathbf{D}^{-1}_{\mathcal{U}\mathcal{U}}\mathbf{A}_{\mathcal{U}\mathcal{U}} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ -\mathbf{W} & \mathbf{V} \end{bmatrix}. \end{aligned} \tag{10}$$

TABLE I

Reference Table for Matrix Shorthand Relationships Used

| Shorthand | Longhand | Dimensions |
|---|---|---|
| $\mathbf{P}$ | $\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} (\mathbf{I} - \mathbf{D}^{-1}\mathbf{A})$ | $\mathbb{R}^{n \times n}$ |
| $\mathbf{Q}$ | $\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} (\mathbf{I} + \mathbf{D}^{-1}\mathbf{A})$ | $\mathbb{R}^{n \times n}$ |
| $\mathbf{U}$ | $\mathbf{I} + \mathbf{D}^{-1}_{\mathcal{U}\mathcal{U}}\mathbf{A}_{\mathcal{U}\mathcal{U}}$ | $\mathbb{R}^{n_{\mathcal{U}} \times n_{\mathcal{U}}}$ |
| $\mathbf{V}$ | $\mathbf{I} - \mathbf{D}^{-1}_{\mathcal{U}\mathcal{U}}\mathbf{A}_{\mathcal{U}\mathcal{U}}$ | $\mathbb{R}^{n_{\mathcal{U}} \times n_{\mathcal{U}}}$ |
| $\mathbf{W}$ | $\mathbf{D}^{-1}_{\mathcal{U}\mathcal{U}}\mathbf{A}_{\mathcal{U}\mathcal{M}}$ | $\mathbb{R}^{n_{\mathcal{U}} \times n_{\mathcal{M}}}$ |

The matrix $\mathbf{D}$ is diagonal and therefore easily inverted. While a graph that has any nodes with degree zero would have a singular $\mathbf{D}$, it is clear from a diffusion perspective, due to these nodes being entirely disconnected from others that propagation of this form would not provide completion for these nodes' features. We therefore would simply exclude any such nodes and therefore guarantee invertibility.

We can now rephrase the optimization problem as follows:

$$\begin{aligned} \hat{h} = \arg\min_{\boldsymbol{h}} \; &||\mathbf{P}\boldsymbol{h}||_2^2 \\ \text{s.t. } &\boldsymbol{h}_{\mathcal{M}} = \boldsymbol{r}_{\mathcal{M}} \end{aligned} \tag{11}$$

which we will show to again have a closed-form solution matching the result given in (7).

Since we can express the objective function as follows:

$$||\mathbf{P}\boldsymbol{h}||_2^2 = ||-\mathbf{W}\boldsymbol{h}_{\mathcal{M}} + \mathbf{V}\boldsymbol{h}_{\mathcal{U}}||_2^2 \tag{12}$$

we can consider stationary points by considering when the derivative with respect to $\boldsymbol{h}_{\mathcal{U}}$ is the zero vector. This is

expressed as follows:

$$\frac{\partial\left(||\mathbf{P}\boldsymbol{h}||_2^2\right)}{\partial \boldsymbol{h}_{\mathcal{U}}} = 2\mathbf{V}^{\mathrm{T}}(-\mathbf{W}\boldsymbol{h}_{\mathcal{M}} + \mathbf{V}\boldsymbol{h}_{\mathcal{U}}) = \mathbf{0}. \tag{13}$$

Considering the block matrix form of the Laplacian

$$\mathbf{L} = \begin{bmatrix} \mathbf{D}_{\mathcal{M}\mathcal{M}} - \mathbf{A}_{\mathcal{M}\mathcal{M}} & -\mathbf{A}_{\mathcal{M}\mathcal{U}} \\ -\mathbf{A}_{\mathcal{U}\mathcal{M}} & \mathbf{D}_{\mathcal{U}\mathcal{U}} - \mathbf{A}_{\mathcal{U}\mathcal{U}} \end{bmatrix} \tag{14}$$

we can substitute $\mathbf{A}_{\mathcal{U}\mathcal{U}}$ for $\mathbf{D}_{\mathcal{U}\mathcal{U}} - \mathbf{L}_{\mathcal{U}\mathcal{U}}$ to obtain a second expression for $\mathbf{V}$, given by

$$\begin{aligned} \mathbf{V} &= \mathbf{I} - \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{A}_{\mathcal{U}\mathcal{U}} \\ &= \mathbf{I} - \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}(\mathbf{D}_{\mathcal{U}\mathcal{U}} - \mathbf{L}_{\mathcal{U}\mathcal{U}}) \\ &= \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{U}}. \end{aligned} \tag{15}$$

By making a similar substitution, we can also obtain a second expression for $\mathbf{W}$. Substitution of $\mathbf{A}_{\mathcal{U}\mathcal{M}}$ for $-\mathbf{L}_{\mathcal{U}\mathcal{M}}$ provides

$$\mathbf{W} = \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{A}_{\mathcal{U}\mathcal{M}} = -\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}. \tag{16}$$

Using these relationships, our optimal $\boldsymbol{h}_{\mathcal{U}}$, given that we must replace $\boldsymbol{h}_{\mathcal{M}}$ with $\boldsymbol{r}_{\mathcal{M}}$ to satisfy our equality constraint, can therefore be found according to

$$\begin{aligned} &2\mathbf{V}^{\mathrm{T}}(-\mathbf{W}\boldsymbol{r}_{\mathcal{M}} + \mathbf{V}\boldsymbol{h}_{\mathcal{U}}) = \mathbf{0} \\ &2\mathbf{V}^{\mathrm{T}}\left(\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}} + \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{U}}\boldsymbol{h}_{\mathcal{U}}\right) = \mathbf{0} \\ &2\mathbf{V}^{\mathrm{T}}\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}(\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}} + \mathbf{L}_{\mathcal{U}\mathcal{U}}\boldsymbol{h}_{\mathcal{U}}) = \mathbf{0}. \end{aligned} \tag{17}$$

From inspection, we can say that when $\boldsymbol{h}_{\mathcal{U}} = -\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}}$, we have a solution, which matches (7).

Letting $\boldsymbol{h}^0$ be the *true* signal, and $\hat{\boldsymbol{h}}$ be our estimation of the signal, we use $\boldsymbol{w} = \boldsymbol{h}^0 - \hat{\boldsymbol{h}}$ to refer to the estimation error. We consider our observed entries to be noiseless and therefore $||\boldsymbol{w}_{\mathcal{M}}||_2 = ||\boldsymbol{r}_{\mathcal{M}} - \hat{\boldsymbol{h}}_{\mathcal{M}}||_2 = 0$. In order to derive an upper bound on the magnitude of $\boldsymbol{w}_{\mathcal{U}}$, the section of the estimation error corresponding to the unobserved nodes, we will make use of two inequalities.

The first inequality is an upper bound on $||\mathbf{P}\boldsymbol{w}||_2$ and is obtained by making use of the triangle inequality and the fact that $||\mathbf{P}\hat{\boldsymbol{h}}||_2 = 0$, as shown in the following equation:

$$\begin{aligned} \left|\left|\mathbf{P}\hat{\boldsymbol{h}}\right|\right|_2 &= ||-\mathbf{W}\boldsymbol{r}_{\mathcal{M}} - \mathbf{V}\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}}||_2 \\ &= \left|\left|-\mathbf{W}\boldsymbol{r}_{\mathcal{M}} - \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{U}}\mathbf{L}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}}\right|\right|_2 \\ &= \left|\left|\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}} - \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{L}_{\mathcal{U}\mathcal{M}}\boldsymbol{r}_{\mathcal{M}}\right|\right|_2 \\ &= 0. \end{aligned} \tag{18}$$

The upper bound on $||\mathbf{P}\boldsymbol{w}||_2$ is given by

$$\begin{aligned} &||\mathbf{P}\boldsymbol{w}||_2 = ||\mathbf{P}\left(\boldsymbol{h}^0 - \hat{\boldsymbol{h}}\right)||_2 \leq ||\mathbf{P}\boldsymbol{h}^0||_2 + \cancelto{0}{||\mathbf{P}\hat{\boldsymbol{h}}||_2} \\ &||\mathbf{P}\boldsymbol{w}||_2 \leq |\psi| \end{aligned} \tag{19}$$

where the parameter $\psi$ represents the cost of the true signal, i.e.,

$$||\mathbf{P}\boldsymbol{h}^0||_2^2 = \psi^2. \tag{20}$$

The second inequality is an upper bound on a conjugate form of the objective function and is presented in the following equation:

$$\begin{aligned} ||\mathbf{Q}\boldsymbol{w}||_2 &= \left|\left|\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{W} & \mathbf{U} \end{bmatrix}\begin{bmatrix} \boldsymbol{w}_{\mathcal{M}} \\ \boldsymbol{w}_{\mathcal{U}} \end{bmatrix}\right|\right|_2 \\ &= ||\mathbf{W}\boldsymbol{w}_{\mathcal{M}} + \mathbf{U}\boldsymbol{w}_{\mathcal{U}}||_2 \\ &= ||\mathbf{U}\boldsymbol{w}_{\mathcal{U}}||_2 \\ &\leq ||\mathbf{U}||_2||\boldsymbol{w}_{\mathcal{U}}||_2 \\ &\leq \phi||\boldsymbol{w}_{\mathcal{U}}||_2. \end{aligned} \tag{21}$$

This bound makes use of a second parameter

$$\phi = ||\mathbf{U}||_2 \tag{22}$$

which is a function of the graph's structure, where $||\cdot||_2$ when applied to a matrix denotes the spectral norm, given by the largest singular value of the matrix.

Using the fact that

$$\begin{aligned} ||(\mathbf{P}+\mathbf{Q})\boldsymbol{w}||_2 &= 2\left|\left|\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}\begin{bmatrix} \boldsymbol{w}_{\mathcal{M}} \\ \boldsymbol{w}_{\mathcal{U}} \end{bmatrix}\right|\right|_2 \\ &= 2||\boldsymbol{w}_{\mathcal{U}}||_2 \end{aligned} \tag{23}$$

we can combine the results of (19) and (21) to obtain

$$\begin{aligned} ||\boldsymbol{w}_{\mathcal{U}}||_2 &= \frac{1}{2}||(\mathbf{P}+\mathbf{Q})\boldsymbol{w}||_2 \\ &\leq \frac{1}{2}(||\mathbf{P}\boldsymbol{w}||_2 + ||\mathbf{Q}\boldsymbol{w}||_2) \\ \therefore ||\boldsymbol{w}_{\mathcal{U}}||_2 &\leq \frac{1}{2}(|\psi| + \phi||\boldsymbol{w}_{\mathcal{U}}||_2). \end{aligned} \tag{24}$$

If $\phi < 2$ then we can rearrange for the final result: the upper bound on the magnitude of the estimation error corresponding to the unobserved nodes, given by

$$||\boldsymbol{w}_{\mathcal{U}}||_2 \leq \frac{|\psi|}{2-\phi}. \tag{25}$$

In terms of when we can say $\phi < 2$, we must consider $||\mathbf{I}+\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{A}_{\mathcal{U}\mathcal{U}}||_2$. Using the disk theorem for eigenvalues [56], we can say that the eigenvalues of $\mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{A}_{\mathcal{U}\mathcal{U}}$ lie in the interval $(-1, 1)$, and therefore, the eigenvalues of $\mathbf{I} + \mathbf{D}_{\mathcal{U}\mathcal{U}}^{-1}\mathbf{A}_{\mathcal{U}\mathcal{U}}$ lie in the interval $(0, 2)$.

### A. Comparison With GTVM Upper Bound

Chen et al. [45] perform the optimization which in our notation is given by

$$\begin{aligned} &\min_{\boldsymbol{h}} \left|\left|\boldsymbol{h} - \frac{1}{|\lambda_{\max}(\mathbf{A})|}\mathbf{A}\boldsymbol{h}\right|\right|_2^2 \\ &\text{s.t. } \boldsymbol{h}_{\mathcal{M}} = \boldsymbol{r}_{\mathcal{M}} \end{aligned} \tag{26}$$

where $\lambda_{\max}(\mathbf{A})$ refers to the largest eigenvalue of the adjacency matrix, as introduced in Section III-A Using $\mathbf{A}' = (1/|\lambda_{\max}(\mathbf{A})|)\mathbf{A}$ to denote the normalized adjacency and under the same noiseless measurement assumptions, their upper bound is given by

$$||\boldsymbol{w}_{\mathcal{U}}||_2 \leq \frac{2|\eta|}{2-q} \tag{27}$$

where $\eta^2 = ||\boldsymbol{h}^0 - \mathbf{A}'\boldsymbol{h}^0||_2^2$ and $q = \left|\left|\begin{smallmatrix} \mathbf{A}'_{\mathcal{M}\mathcal{U}} \\ \mathbf{I}+\mathbf{A}'_{\mathcal{U}\mathcal{U}} \end{smallmatrix}\right|\right|_2$. Their $q$ lies in the interval $(0, 2)$ and is analogous to the quantity $\phi$ in our bound.

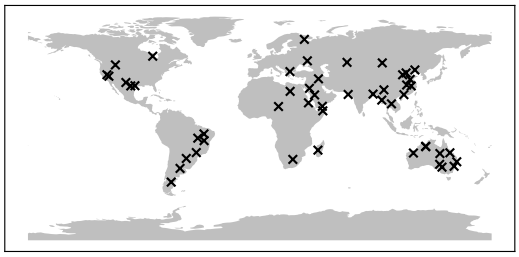


Fig. 2. Locations of 50 randomly sampled locations for which Landsat 7 acquisitions have been acquired. For each location, a pair of cloud-free images, acquired between the start of 2001 and the end of 2002, has been obtained.

## V. Experimental Results and Analysis

In this section, the graph-based propagation method is compared against existing methods for image completion. To do so, we perform a set of simulated experiments in which image regions are artificially removed and each method is used to complete the image and their results compared. In advance of these results, we first provide an outline of the remote sensing dataset which we have used and details about the simulated removal of data which has been performed.

The code used for the experiments presented in this work has been made publicly available at: https://github.com/iainrolland/GraphProp.

### A. Dataset

In order to quantitatively benchmark each method, a dataset of remote sensing images which is without missing data was prepared. The dataset, which consists of multispectral Landsat 7 acquisitions, covers 50 randomly selected locations across the globe, as shown in Fig. 2. At each location, the images acquired by the Landsat 7 platform in the years 2001 and 2002 were downloaded and manually filtered to obtain a cloud-free pair of acquisitions at each location. The result is a dataset consisting of 100 images, two at each location, each image consisting of seven spectral bands. The spatial dimensions of the images range from 418 to 429 pixels in height and from 183 to 418 pixels in width.

To validate our approach, we simulate the missing data by artificially removing image sections and comparing the quality each method's completion against the original images. The two simulated experiments were designed so as to resemble real-world problems encountered in remote sensing.

### B. Experiments

*1) Simulated SLC-Off Experiment:* The first experiment replicates a common reconstruction problem which came about due to a sensor failure on board the Landsat 7 satellite in 2003. As a result of the scan line corrector-off (SLC-off) problem, the satellite acquires data with bands of missing pixels [3]. In this experiment, the SLC-off problem is simulated by introducing deadlines in the latter image of each pair of Landsat 7 images. The SLC-off problem causes missing information in all of the image bands and so this is replicated in how data has been removed in this simulated experiment. The former acquisition of each location pair is provided without missing data and thus provides valuable information in the completion task.

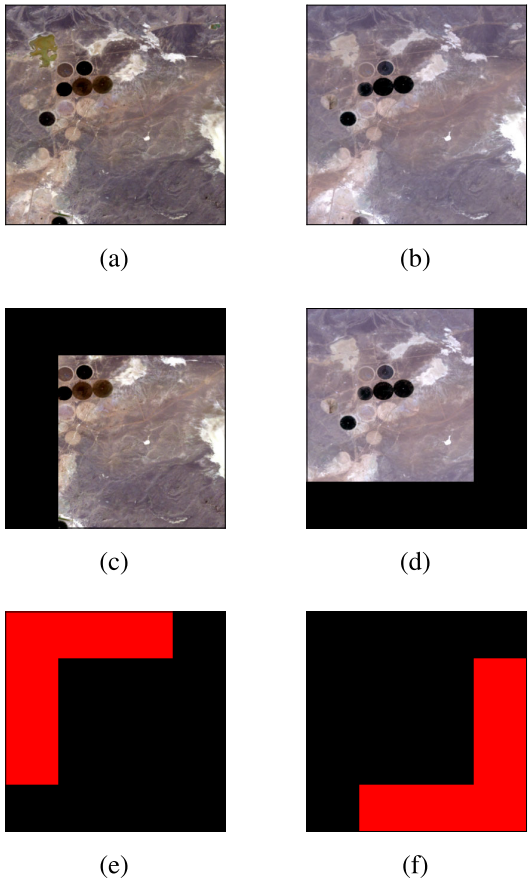



Fig. 3. Simulated partial overlap experiment. The full images are shown in (a) and (b) representing the first and second acquisitions, respectively. The images provided representing a partial overlap experiment, where 40% of the original area has been removed are shown in (c) and (d) taken from data in (a) and (b), respectively. The regions which are to be completed and over which metrics are computed are shown for (c) and (d) in red in (e) and (f), respectively.

*2) Simulated Partial Overlap:* In the second experiment, we consider a case where two acquisitions have been made which overlap only partially. This represents the common case where satellite orbits have followed different paths and therefore have provided data which do not perfectly align. The goal of this setup is to provide a completed image in the regions which were observed in only one image and therefore lie out with the region of overlap. This problem, to the best of the authors' knowledge, has not been studied previously. To simulate this scenario we take each location's Landsat 7 image pair in the aforementioned dataset and remove a fixed number of rows and columns from the left and top of the first-acquired image and the same fixed number of rows and columns from the right and bottom of the second image. In testing, the number of rows and columns removed has been varied in order to explore how each method performs as the extent of overlap is varied. Since the exact shape of data at each of the 50 locations varies slightly, the amount of data removed is expressed as the area removed as a percentage of the full image area. An example of how the partial overlap experiment is simulated is shown in Fig. 3.

### C. Experimental Results

Benchmarks have been provided to compare our approach to state-of-the-art methods in both low-rank completion as well as graph-based signal recovery. The following benchmarks are all based on some form of low-rank approach to image completion: high accuracy low rank tensor completion (HaLRTC) [27]; adaptive weighted tensor completion (AWTC) [16]; multigraph-based low-rank-tensor-approximation (MGLRTA) [29]; and total variation regularized weighted tensor ring decomposition (TVWTR) [30]. An additional benchmark from graph signal recovery literature is provided by GTVM [45]. In all tests, benchmarks have been implemented using the author-recommended parameter settings where supplied. Where recommended parameters are not given or a range of values is recommended, parameters were selected following a tuning process to produce the best results on the aforementioned dataset.

The low-rank completion approaches have been applied by stacking the two images to form a fourth-order tensor (spatial–spatial–spectral–temporal). In the graph-based methods, the observed regions are exploited to construct the structure of the graph. More details as to how this was performed for each simulated experiment are given in Sections V-C1 and V-C2, respectively.

To quantitatively compare the quality of the recovery, we compute four metrics to evaluate the performance of each method. These are the mean squared error (MSE), root MSE (RMSE), mean absolute error (MAE), and mean peak signal-to-noise ratio (mPSNR), and they are defined as follows:

$$\text{MSE} = \frac{1}{n_{\mathcal{U}}}||\boldsymbol{w}_{\mathcal{U}}||_2^2 \tag{28}$$

$$\text{RMSE} = \frac{1}{n_{\mathcal{U}}}||\boldsymbol{w}_{\mathcal{U}}||_2 \tag{29}$$

$$\text{MAE} = \frac{1}{n_{\mathcal{U}}}||\boldsymbol{w}_{\mathcal{U}}||_1 \tag{30}$$

$$\text{PSNR} = 10\log_{10}\left(\frac{\max_{\boldsymbol{h}}^2}{\frac{1}{n_{\mathcal{U}}}||\boldsymbol{w}_{\mathcal{U}}||_2^2}\right) \tag{31}$$

where $||\cdot||_2$, $||\cdot||_1$, and $\max_{\boldsymbol{h}}$ denote the $l_2$ norm, $l_1$ norm, and the maximum possible pixel value in the band, respectively. In each case, these have been computed as an average over all the spectral bands over the missing regions of the image only. Where a pixel was removed from both acquisitions these were also excluded from the metric as we prioritize the quality of recovery in the areas for which a reasonable predicted recovery can be made. The PSNR values are given as the mean of the PSNR value for each band, named mPSNR.

*1) Simulated SLC-Off Experiment:* For each image pair, the first acquisition was provided without missing data. For the graph-based methods, this acts as the reference acquisition from which the graph is obtained. By computing the $k$-nearest neighbors of each node, the graph is constructed so as to connect any two nodes which belong to either node's $k$-nearest neighbors. In all investigations, $k = 20$ has been used. This graph is then used to propagate the data with the simulated SLC-off issue. For the low-rank completion methods, the reference image is stacked with the simulated SLC-off image to obtain a fourth-order tensor which is to be completed.

TABLE II

SIMULATED SLC-OFF EXPERIMENT. VALUES IN MAGENTA AND BLUE SHOW THE BEST AND SECOND-BEST METHOD, RESPECTIVELY

| Method | MSE | RMSE | MAE | mPSNR (dB) |
|---|---|---|---|---|
| AWTC | 0.0293 | 0.1568 | 0.1153 | 16.8162 |
| HaLRTC | 0.0095 | 0.0944 | 0.0687 | 20.8393 |
| GTVM | 0.0587 | 0.2050 | 0.1669 | 15.1486 |
| MGLRTA | 0.0237 | 0.1516 | 0.1202 | 16.5415 |
| TVWTR | 0.0101 | 0.0920 | 0.0666 | 21.4597 |
| GraphProp | 0.0059 | 0.0715 | 0.0502 | 23.6713 |

TABLE III

SIMULATED PARTIAL OVERLAP EXPERIMENT FOR THE CASE WHERE 40% OF THE AREA HAS BEEN REMOVED. VALUES IN MAGENTA AND BLUE SHOW THE BEST AND SECOND-BEST METHOD, RESPECTIVELY

| Method | MSE | RMSE | MAE | mPSNR (dB) |
|---|---|---|---|---|
| AWTC | 0.0628 | 0.2393 | 0.1850 | 12.8305 |
| HaLRTC | 0.0364 | 0.1873 | 0.1423 | 14.7102 |
| GTVM | 0.1573 | 0.3927 | 0.3310 | 8.2061 |
| MGLRTA | 0.0657 | 0.2500 | 0.2038 | 12.2491 |
| TVWTR | 0.0616 | 0.2406 | 0.1824 | 12.6523 |
| GraphProp | 0.0103 | 0.0966 | 0.0656 | 20.7907 |

Table II presents the results of the simulated SLC-off experiment as measured by each of the four metrics considered. Tabulated values represent averages computed across the 50 locations in the Landsat 7 dataset. The values in Table II show a significantly improved image completion is obtained using GraphProp than with even the next-best performance obtained using TVWTR. The improvement is even more significant when compared with either other benchmarks. A visual representation of these results is presented in Fig. 4. These results depict obvious artifacts of the deadlines in the AWTC, MGLRTA, and to a lesser extent in the HaLRTC completions. The GTVM method handles this scene successfully with less obvious deadline artifacts but has visible amounts of noise in the completed regions. The TVWTR results are also impressive with only the artifacts of deadlines only subtly visible. The GraphProp method provides results which are visibly superior than those obtained using the benchmarked methods. Although still present, the deadline artifacts are the least obvious and it does not suffer from the same noisy completion as seen in the GTVM completion.

*2) Simulated Partial Overlap:* In this experiment, a graph was produced using the visible pixels of the first image and combined with a graph computed using the observed pixels in the second image. The graphs are combined by taking each graph's edge set and creating the combined graph using a logical OR operation between the edge sets, such that the graph remains unweighted. This graph adjacency is the same for GTVM and GraphProp. The low-rank completion approaches have again been applied to the tensor obtained by stacking the two third-order tensors in the fourth dimension.

The performance of each method at 40% area removed can be found from Table III. These values represent the values obtained by averaging performance across the 50 locations. Once more the GraphProp completion is shown to obtain an improved-quality reconstruction. In this case, the difference between the quality of the GraphProp completion and the

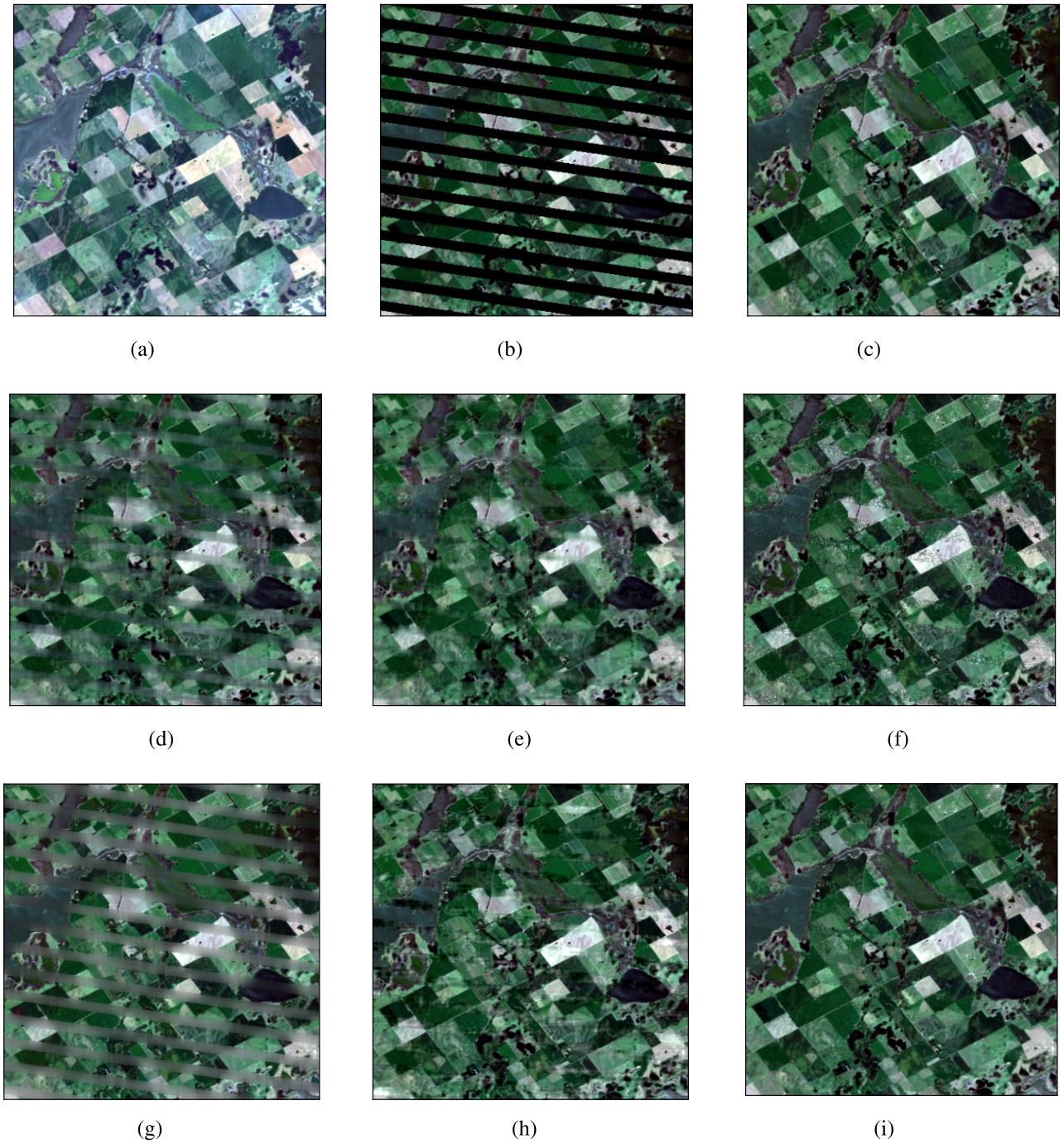



Fig. 4. Simulated SLC-off experiment where Day A is fully observed and Day B is obscured with simulated SLC-off pattern. (a) Observation Day A. (b) Observation Day B. (c) Ground Truth Day B. (d) AWTC Day B. (e) HaLRTC Day B. (f) GTVM Day B. (g) MGLRTA Day B. (h) TVWTR Day B. (i) GraphProp Day B.

benchmark methods is even more significant than was shown in Table II.

In particular, the GTVM method scores very poorly at this task. One theory for why GraphProp outperforms GTVM so significantly relates to how the GTVM method normalizes the adjacency matrix within the total variation objective function. The definition of graph total variation [57] uses a normalized adjacency matrix obtained by dividing by the eigenvalue of largest magnitude. For graph nodes which have a degree much smaller than the largest magnitude eigenvalue, the result is a row in the normalized adjacency which sums to much less than 1. The total variation minimization therefore shows the preference toward solutions that fill the corresponding missing entries with smaller magnitude values than the magnitude of observed entries. Subsequently presented visualisations of the GTVM output also support this theory, with reconstructed missing values often visually darker than the rest of the image pixels.

Of the tested low-rank completion methods the HaLRTC algorithm handles this scenario best. All methods, however, show significantly inferior reconstruction ability for this task when compared against GraphProp. Dissecting the results into individual locations, as displayed in Fig. 5, demonstrates the consistency of the GraphProp approach, with the

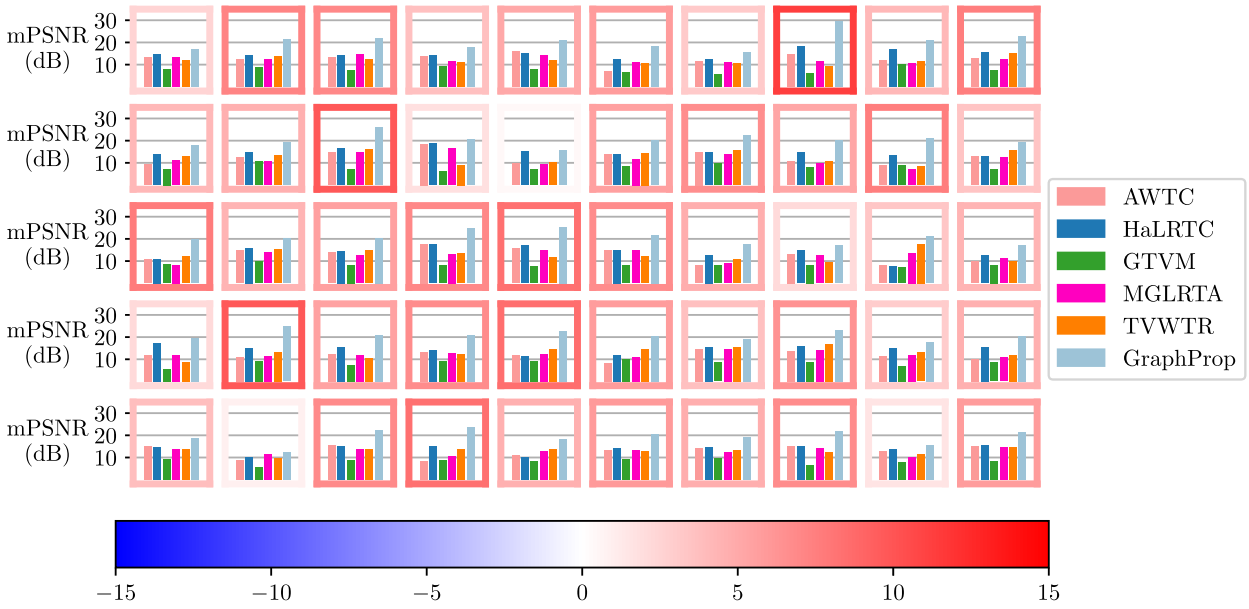



Fig. 5. For each of the 50 locations the computed mPSNR for each method's completion in the partial overlap experiment with 40% area removed. The border of each bar chart is colored according to the horizontal gradient and represents the mPSNR obtained using GraphProp minus the mPSNR of the best benchmarked method.

method providing the highest mPSNR among the tested methods not only on average but also at all of the 50 tested locations.

A visual representation of performance for this experiment has been provided in Fig. 6. The two acquisitions differ slightly in terms of their illumination and coloring which allows us to assess each method's robustness to ground surface changes.

In this figure, we see in AWTC results a tendency to almost directly transfer values across the temporal dimension. This provides a reconstruction which suffers from sharp edge artifacts and color discontinuities. The HaLRTC algorithm much better handles the temporal differences with a completion that more closely matches the spectral appearance of the ground-truth scene. It does, however, suffer from horizontal and vertical banding. This is likely a direct consequence of the low-rank assumption, with the optimization favoring solutions consisting of similar rows and columns.

The MGLRTA approach provides blurrier completed entries than other tested methods. Although blurry, the ground patterns can be faintly observed in the MGLRTA solutions. In these partial overlap results, the TVWTR algorithm exhibits similar banding issues to the HaLRTC solution but with even greater severity. The poor performance of the low-rank methods is unsurprising and likely relates to earlier discussion which states that these methods were not designed to handle large contiguous blocks of missing entries [19].

The visual inspection of the GTVM output confirms, as was shown in Table III, that the GTVM method produces the lowest quality reconstruction. The missing regions are completed with very low-brightness resemblances of the original scene, likely explained by the method's use of the normalized adjacency matrix. The GraphProp method, however, completes the missing regions well. It handles the change in light and color between the two acquisitions and shows only very slight signs of edge artifacts between the observed and completed sections.

As an extension to the partial overlap experiment, an investigation which considers how each method handles varying amounts of overlap between the two images was conducted. By varying the amount of area removed between 10% and 70%, we show performance as a function of overlap extent. Tests with beyond 70% area removed were not conducted as they would represent a scenario in which no overlap between the two images was given. The performance as measured by mPSNR for this test is presented in Fig. 7. This figure shows that GraphProp achieves a higher mPSNR reconstruction across all scenarios. Although there is an accelerated deterioration in mPSNR beyond 60% area removed, the GraphProp approach still achieves results at 70% area removed which are approximately comparable with the mPSNR provided by the best benchmarked methods at only 10% area removed.

A qualitative comparison of results at 70% area removed is provided in the Appendix.

### *D. Ablation Study*

In this section, we present two studies: an experiment which validates the use of $k$-nearest neighbors as the chosen method for graph construction; and results showing that diffusion-based propagation performs favorably against other graph signal completion approaches.

*1) Graph Construction Validation:* To validate GraphProp's use of a $k$-nearest neighbors graph as opposed to alternative methods of graph construction, the comparison has been made against results obtained by instead using a radius neighbors graph. The radius neighbors graph, denoted here by RadGraph, connects two nodes with an edge if the node features are within a specified Euclidean distance from one another. To isolate the impact of using RadGraph over a

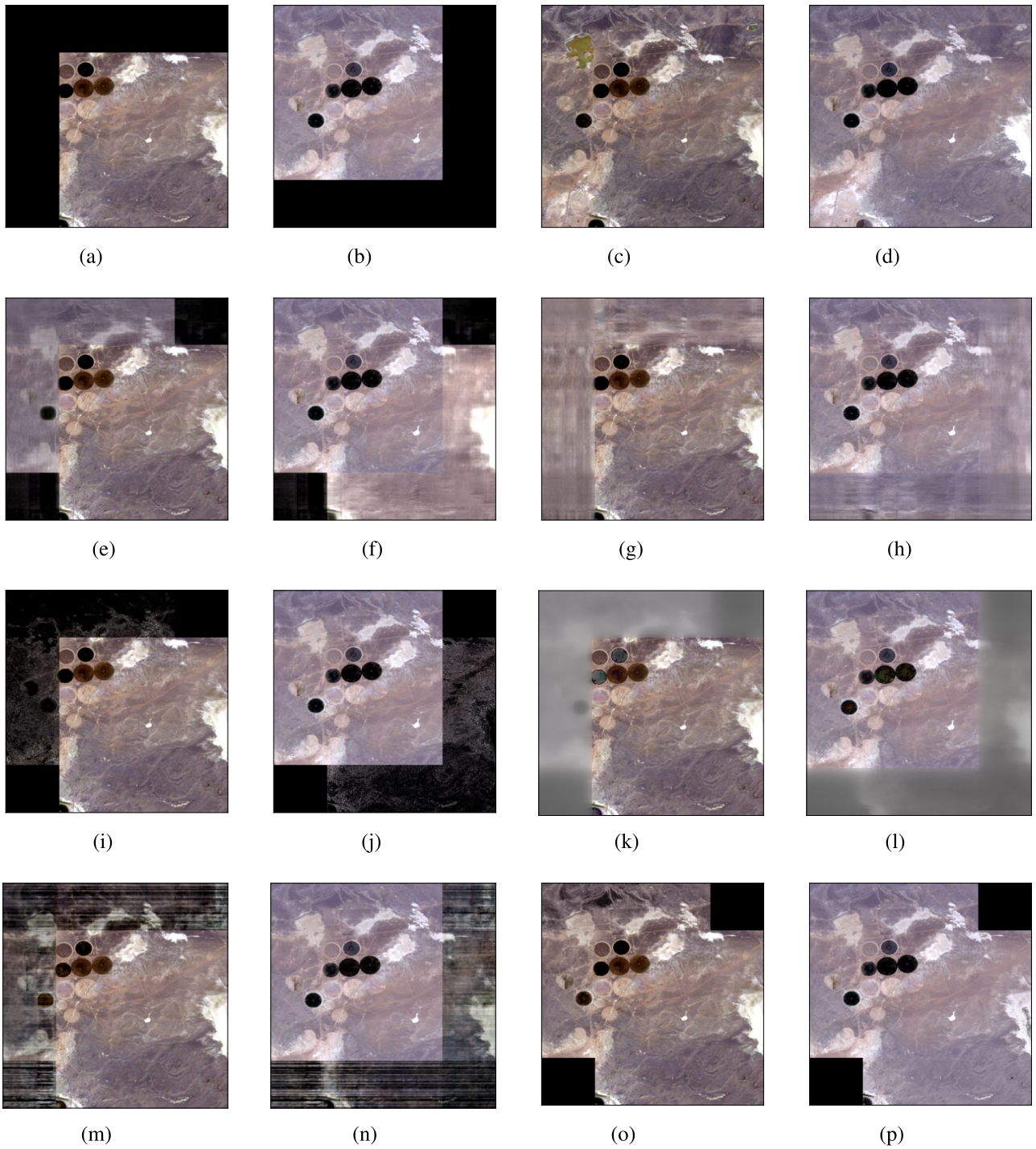



Fig. 6. Visual depiction of results from the simulated partial overlap experiment with 40% area removed. The first and third columns represent the first acquisition, while the second and fourth columns depict the second acquisition. (a) Observation Day A. (b) Observation Day B. (c) Ground-truth Day A. (d) Ground-truth Day B. (e) AWTC Day A. (f) AWTC Day B. (g) HaLRTC Day A. (h) HaLRTC Day B. (i) GTVM Day A. (j) GTVM Day B. (k) MGLRTA Day A. (l) MGLRTA Day B. (m) TVWTR Day A. (n) TVWTR Day B. (o) GraphProp Day A. (p) GraphProp Day B.

$k$-nearest neighbors graph, the ablation experiment maintains the same diffusion-based propagation used by GraphProp to perform completion. To indicate this, we use "RadGraph + Diffusion" to refer to this method. The radius threshold chosen was 0.05 which was selected after testing a series of trial radii.

*2) Graph Signal Completion Validation:* To justify the use of diffusion-based propagation in GraphProp versus alternative graph signal completion methods comparison has also been made against graph nearest-neighbor interpolation (GraphNearestN). Instead of propagating observations along graph edges, the GraphNearestN, fills missing node features with the node feature from the nearest observed node by number of graph hops. In these results, the same $k$-nearest neighbors graph as GraphProp is used. Only the completion step has been substituted. To indicate this, "kNN + GraphNearestN" is used to refer to this method.

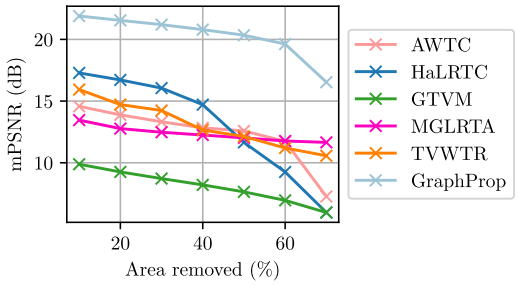



Fig. 7. Performance of each method in the partial overlap experiment as the percentage of area removed is varied. The plotted series represent the averages obtained across the 50 locations tested.

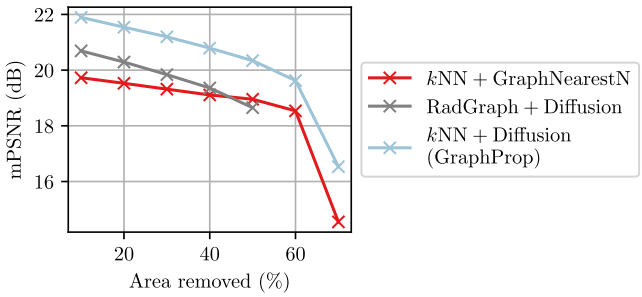



Fig. 8. Performance of ablation methods in the partial overlap experiment as the percentage of area removed is varied. The plotted series represents the averages obtained across the 50 locations tested.

TABLE IV

ABLATION STUDY USING SIMULATED SLC-OFF EXPERIMENT. VALUES SHOWN REPRESENT THE MEAN ± STANDARD DEVIATION

| Method | mPSNR (dB) | Runtime (s) |
|---|---|---|
| $k$NN +GraphNearestN | $20.7488 \pm 3.7551$ | $1.7323 \pm 0.1189$ |
| RadiusGraph +Diffusion | $20.0896 \pm 2.5200$ | $495.5588 \pm 882.6159$ |
| $k$NN+Diffusion (GraphProp) | $23.6713 \pm 3.8090$ | $11.2697 \pm 7.2106$ |

TABLE V

ABLATION STUDY USING SIMULATED PARTIAL OVERLAP EXPERIMENT WITH 40% OF AREA REMOVED. VALUES SHOWN REPRESENT THE MEAN ± STANDARD DEVIATION

| Method | mPSNR (dB) | Runtime (s) |
|---|---|---|
| $k$NN +GraphNearestN | $19.1085 \pm 3.3075$ | $73.0722 \pm 16.1346$ |
| RadiusGraph +Diffusion | $19.3590 \pm 2.4908$ | $2809.1344 \pm 6215.3348$ |
| $k$NN+Diffusion (GraphProp) | $20.7907 \pm 3.0101$ | $83.6733 \pm 20.8535$ |

In the presented ablation study, GraphProp is represented by the combination "kNN + Diffusion," where a $k$-nearest neighbors graph has been used, it has been obtained using $k = 20$ neighbors.

In Tables IV and V, the results of the ablation study are presented for both the simulated SLC-off and partial overlap experiment (where 40% of the area has been removed), respectively. These results show that the GraphProp combination of a $k$-nearest neighbors graph with diffusion-based propagation completion is the combination which provides the highest accuracy completion. While the other tested combinations are of lower accuracy, the drop-off is not significant. For example, each mPSNR in Table V is higher than the best benchmark in Table III. This suggests the choice to take a graph-based approach to completion approach is sensible and that results are robust to the specific implementations of such an approach. These tables also indicate that for an application scenario which has a greater emphasis on low runtimes and can compromise slightly on accuracy, the $k$NN + GraphNearestN combination might be chosen over the diffusion-based completion of GraphProp.

In Fig. 8, the ablation study is presented on the partial overlap experiment with performance as a function of area removed shown. The presented results corroborate the conclusions of the previous paragraph, where the GraphProp combination provides the highest accuracy results across all area removed percentages. Each of the tested ablation combinations outperforms benchmarked methods plotted in Fig. 7.

The "RadiusGraph + Diffusion" combination is tested only between 10% and 50% area removed due to its increased computational demands. Using the RadiusGraph method is more computationally intensive due to the graph generally consisting of many additional edges. Although using a smaller radius would reduce the number of edges in the graph, it also increases the likelihood of some nodes becoming entirely disconnected in the graph. The $k$-nearest neighbors graph does not have this same issue as it creates a graph, where all nodes have generally similar degrees, $\geq k$.

### E. Impact of Number of Nearest Neighbors

One of the major advantages of the proposed GraphProp completion algorithm is that is has only one single parameter: the number of nearest neighbors, $k$, used to obtain the graph. The results until this point have made use of $k = 20$, which the authors recommend as a sensible suggestion for the parameter setting. However, in this section, we present a study which has been performed in order to provide insight into how performance varies as a function of $k$. In these tests, we have considered $k \in \{2, 6, 10, 20, 40, 60, 100\}$. We do so for both the simulated SLC-off as well as the simulated partial overlap experiments.

*1) Simulated SLC-Off Experiment:* In Fig. 9, the SLC-off experiment has been repeated for each of the aforementioned values of $k$ across the 50 dataset locations and the mean and standard deviation of the mPSNR computed for these results.

*2) Simulated Partial Overlap:* In Figs. 11 and 12, we repeat the accuracy and runtime analysis, respectively, for the partial overlap experiment. In this case, the results are grouped by the percentage of area removed. These results provide additional evidence to suggest $k = 20$ provides fast and near-optimal accuracy at all tested percentages of area missing.

In this figure, we see that performance improves with larger $k$ until a maximum occurs at $k = 20$. For $k$ much larger than this there is a slight deterioration in performance but no strong drop-off occurs. The fact that near-optimal accuracy can be achieved over such a wide range of $k$ is evidence of the robustness of the GraphProp approach.

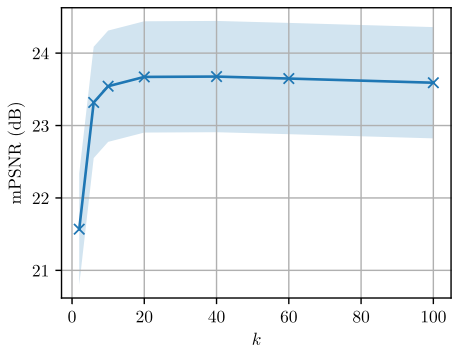



Fig. 9. GraphProp completion accuracy in the SLC-off experiment as a function of the number of nearest neighbors, $k$, used to obtain the graph. The plotted line represents the mean across the 50 locations with the shaded interval representing $\pm$ one standard deviation.

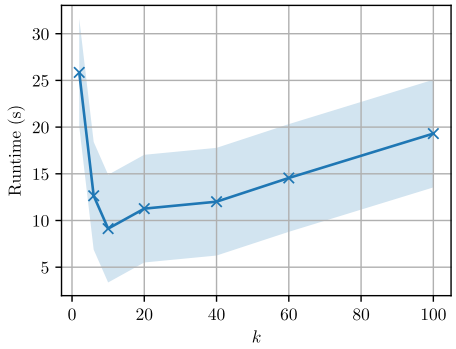



Fig. 10. GraphProp runtime in the SLC-off experiment as a function of the number of nearest neighbors, $k$, used to obtain the graph. The plotted line represents the mean across the 50 locations with the shaded interval representing $\pm$ one standard deviation.

If we consider the runtimes plotted in Fig. 10, we see that there is a computational cost associated with setting $k$ to be too large. In presented experiments, we have used $k = 20$ as it was considered to provide the optimal runtime-accuracy tradeoff.

### *F. Impact of Graph Similarity Across Acquisitions*

Given that the GraphProp method completes the missing entries by relying on graph edges which were constructed using an observation of the region acquired on a different date, it is worth analyzing to what extent performance is affected by how closely the acquisitions resemble one another.

In the dataset described in Section V-A, the temporal separation between each location's image pair varies from between 7 days and 288 days, with a mean separation of 63.7 days. Different regions and land uses are likely to change according to the dynamics of varying timescales. Therefore, rather than presenting completion accuracy as a function of temporal separation, scene similarity has been measured by considering the respective $k$-nearest neighbors graphs. Using the ground-truth acquisitions (without values removed) we obtain the $k$-neighbors graph for each acquisition separately and measure the distance between the respective edge sets.

To quantify graph similarity we use the Jaccard index [58], which expresses as a value between 0 and 1 how different or similar two sets are, respectively. The Jaccard index is

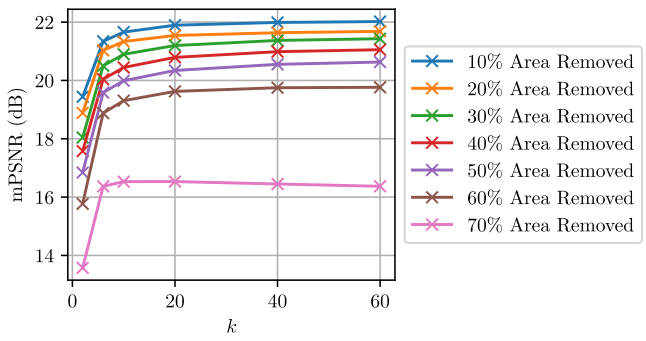



Fig. 11. GraphProp completion accuracy in the partial overlap experiment as a function of the number of nearest neighbors, $k$, used to obtain the graph. Plotted lines represent the mean across the 50 locations for a given percentage of area removed.

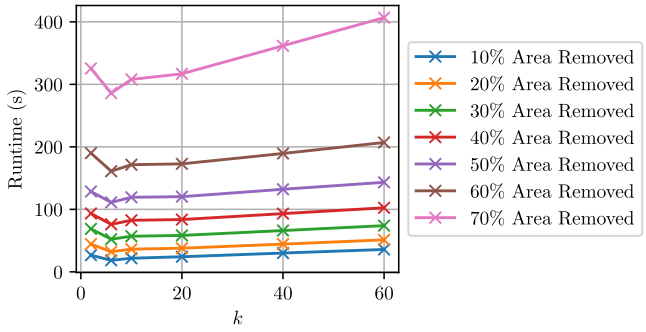



Fig. 12. GraphProp runtime in the partial overlap experiment as a function of the number of nearest neighbors, $k$, used to obtain the graph. Plotted lines represent the mean across the 50 locations for a given percentage of area removed.

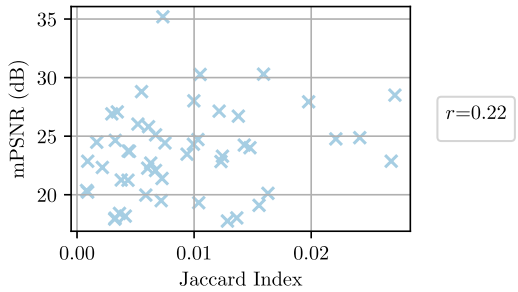



Fig. 13. GraphProp completion accuracy in the SLC-off experiment as a function of the Jaccard index measuring similarity between $k$-nearest neighbors graph edge sets. Each point plotted represents one of the 50 tested locations. The value $r$ represents the Pearson correlation coefficient.

computed as the ratio between the size of the two sets' intersection divided by the size of the sets' union. In this article, an edge is considered in the union of the graphs if an edge connects node $i$ with node $j$ in both graphs.

To study the correlation between GraphProp performance and the scene similarity, the Jaccard index has been computed for each of the 50 dataset locations and plotted against GraphProp accuracy given by the scene's mPSNR value. From these plotted points, the Pearson correlation coefficient has been computed.

In Fig. 13, depicting the SLC-off experiment, we see only a weak positive correlation between the Jaccard index and the achieved mPSNR. This figure, therefore, only weakly suggests

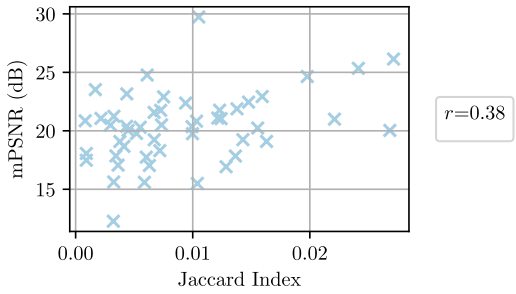



Fig. 14. GraphProp completion accuracy in the partial overlap experiment as a function of the Jaccard index measuring similarity between $k$-nearest neighbors graph edge sets. Each point plotted represents one of the 50 tested locations. The value $r$ represents the Pearson correlation coefficient.

TABLE VI
COMPUTATION TIMES (SLC-OFF EXPERIMENT)

| Method | Average runtime (s) |
|---|---|
| AWTC | 258.8 |
| HaLRTC | 234.9 |
| GTVM | 17.4 |
| MGLRTA | 41.5 |
| TVWTR | 285.8 |
| GraphProp | 16.3 |

TABLE VII
COMPUTATION TIMES (PARTIAL OVERLAP EXPERIMENT WITH 40% AREA REMOVED)

| Method | Average runtime (s) |
|---|---|
| AWTC | 261.2 |
| HaLRTC | 225.0 |
| GTVM | 118.9 |
| MGLRTA | 40.6 |
| TVWTR | 289.5 |
| GraphProp | 137.2 |

that GraphProp performance deteriorates when scene similarity is low between the acquisition pair. Equivalently, we see that even in locations where the two acquired observations exhibit low scene similarity (as indicated by a low Jaccard index), that near-optimal mPSNR values can be obtained.

In the equivalent results for the partial overlap experiment, as shown in Fig. 14, a slightly stronger positive correlation is presented. This shows, therefore, that GraphProp performs marginally better when scene similarity is high. However, when scene similarity is low, the compromise in performance is modest.

### G. Computational Complexity Study

In addition to a comparison of the quality of the completions, we also provide a comparison of the computation times of each method in each experiment. In each case, the experiment has been performed on a desktop computer with an Intel Core i9-10900X CPU and 32 GB of RAM. The MGLRTA and TVWTR algorithms were performed using MATLAB with the remaining methods implemented in Python.

Tables VI and VII show the runtime for each method when applied to the SLC-off and partial overlap experiment with 40% area removed, respectively.

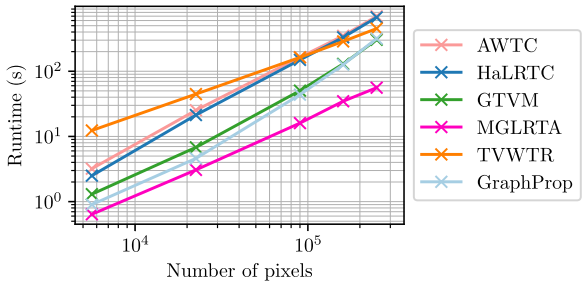



Fig. 15. Time taken to perform the SLC-off experiment. Plotted values represent the mean value across the dataset given as a function of the number of total image pixels. Both axes use a logarithmic scale.

In the SLC-off experiment, the two graph-based methods (GraphProp and GTVM) are faster than all other tested benchmarks. The runtimes are approximately half that of MGLRTA and an order of magnitude faster than all other low-rank completion methods. The GraphProp runtime represents the entire process including the time taken to compute the $k$-nearest neighbors graph.

In the partial overlap problem, GraphProp runtimes are higher than in the SLC-off experiment as it requires a $k$-nearest neighbors graph to be computed for both acquisitions and then an additional overhead involved with finding a solution for each acquisition using this graph. From the benchmarks, both MGLRTA and GTVM produce results faster than GraphProp. The results in Table III, however, show that although they are faster, MGLRTA and GTVM provided the poorest scoring results across all metrics. GraphProp, however, produces significantly better results with runtimes only marginally higher than GTVM and almost a factor of 2 faster than AWTC, HaLRTC, and TVWTR.

To analyze runtimes as a function of the image size, an additional set of tests was performed in the SLC-off context using cropped and resized dataset acquisitions. This allowed each method to be tested when given square inputs having spatial height and width of 75, 150, 300, 400, and 500 pixels. Therefore each method was tested on images containing 5625, 22 500, 90 000, 160 000 and 250 000 pixels, respectively. The runtime results can be seen from Fig. 15, with the runtime in seconds plotted against number of image pixels using a logarithmic scale on both axes.

A straight line in Fig. 15 would represent a power law relationship, and therefore, a method of $\mathcal{O}(n^{\delta})$ complexity, where $n$ is the number of pixels. The power law exponent, $\delta$, can be obtained by finding each straight line's gradient. The plotted results show that GraphProp complexity scales similar to benchmarked approaches. In large-image applications, for example, images consisting of more than say $1 \times 10^6$ pixels, similar quality results could likely be obtained by applying GraphProp in a patchwise fashion. By representing the $1 \times 10^6$ pixels as four regions consisting of $2.5 \times 10^5$ pixels, lower runtimes could be achieved. An investigation which considers when patchwise application of GraphProp becomes preferable and assesses accuracy tradeoffs is outside the scope of this particular work but is mentioned as a possible direction for future research.

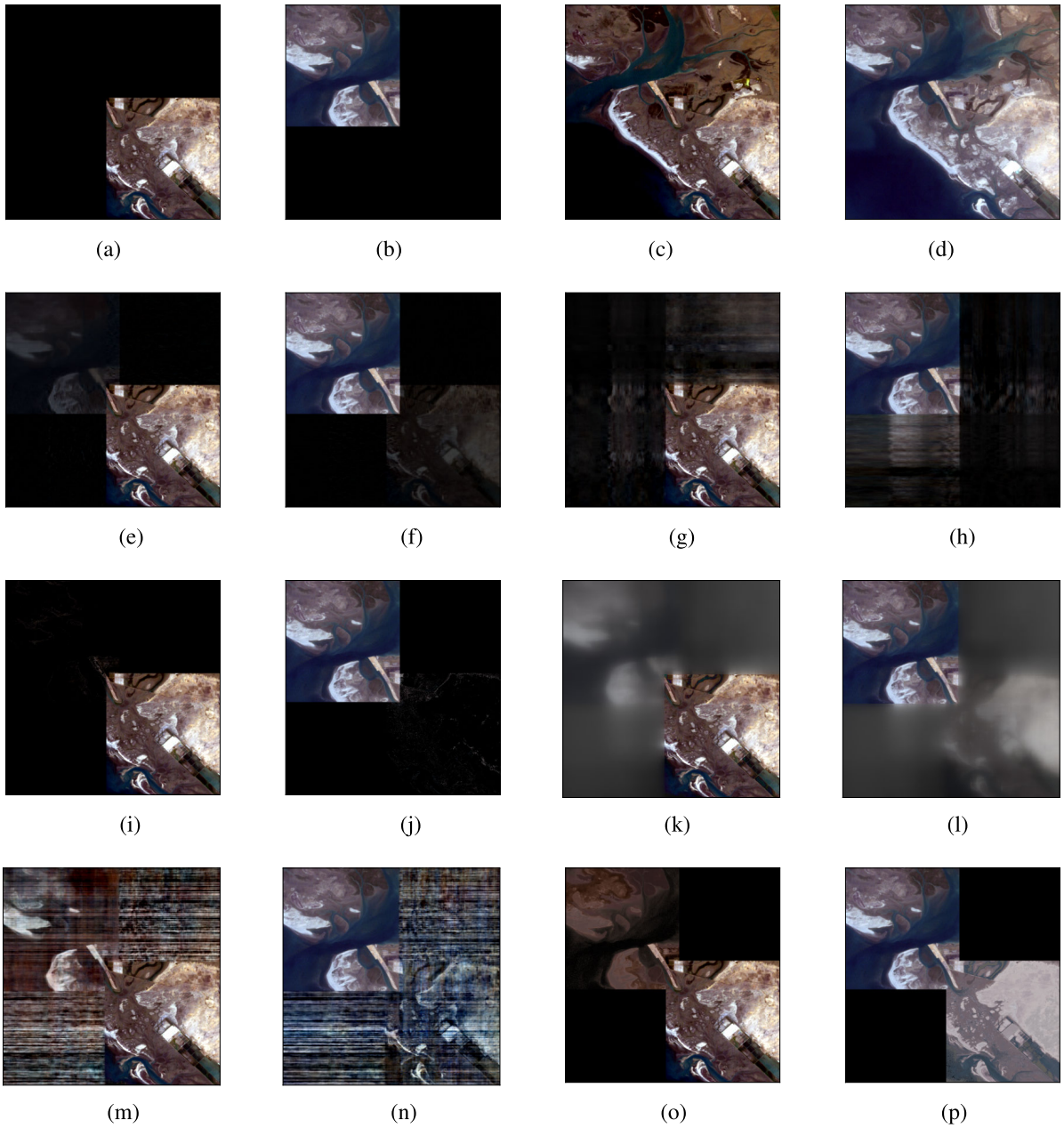



Fig. 16. Extreme case test of partial overlap, where 70% of both images has been removed leaving only 0.8% of the spatial area observed in both images. The provided inputs are shown in (a) and (b) representing the first and second acquisitions, respectively. The ground truth is depicted in (c) and (d). In (e) and (f), the AWTC completion is shown. In (g) and (h), HaLRTC results are shown. The GTVM completion is depicted in (i) and (j). In (k) and (l), the GraphProp completion is shown.

## VI. Conclusion

In this article, we have presented a novel remote sensing image completion method, derived from an analogy to heat-based diffusion. The graph-based propagation approach constructs a graph based on spectral similarity and is used to propagate observed values around the graph in order to obtain a completed image. The proposed method has been proven experimentally to be robust in challenging image completion scenarios in the domain of remote sensing. Experimental results show the approach to outperform the state-of-the-art both in terms of low-rank completion methods and in terms of graph signal recovery methods. In addition to the improved performance, we have also shown that the proposed method is computationally efficient and can be performed quicker than many of the benchmarked low-rank tensor completion algorithms and comparably with other graph signal recovery methods.

There is scope for further research to consider extended applications of the approach. For example, the future work might investigate the multimodal abilities of the proposed method. Investigations which consider how performance is affected when the reference image is acquired using one sensor, e.g., SAR, and the obscured image is acquired using another sensor, e.g., optical, would be informative. In addition, the future study might consider the generalization of this approach to time series of greater than two images. The method could be extended to do so by either: combining

graphs using the union of edge sets obtained from each acquisition (equivalent to the logical OR approach used in the partial overlap experiment) or by creating the graph adjacency from the sum over each acquisition's adjacency matrix, and allowing a weighted adjacency matrix to be used in the diffusion process.

## APPENDIX

Fig. 16 shows an additional qualitative analysis of results in the partial overlap experiment. The depicted test represents the 70% area removed context and shows that GraphProp is robust to this extreme scenario. In this article, the overlap between acquisitions represents less than 1% of total area.

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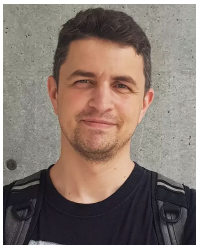
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